

Introduction

- Everyone of us is surrounded by / with materials
- We live in a world that is both dependent upon and limited by materials.
- Everything we see and use is made of materials: cars, airplanes, computers, refrigerators, microwave ovens, TVs, dishes, silverware, athletic equipment of all types, and even biomedical devices such as replacement joints and limbs.
- All of these require materials specifically tailored for their application
- World is materialistic, You can't imagine a world without materials!!!

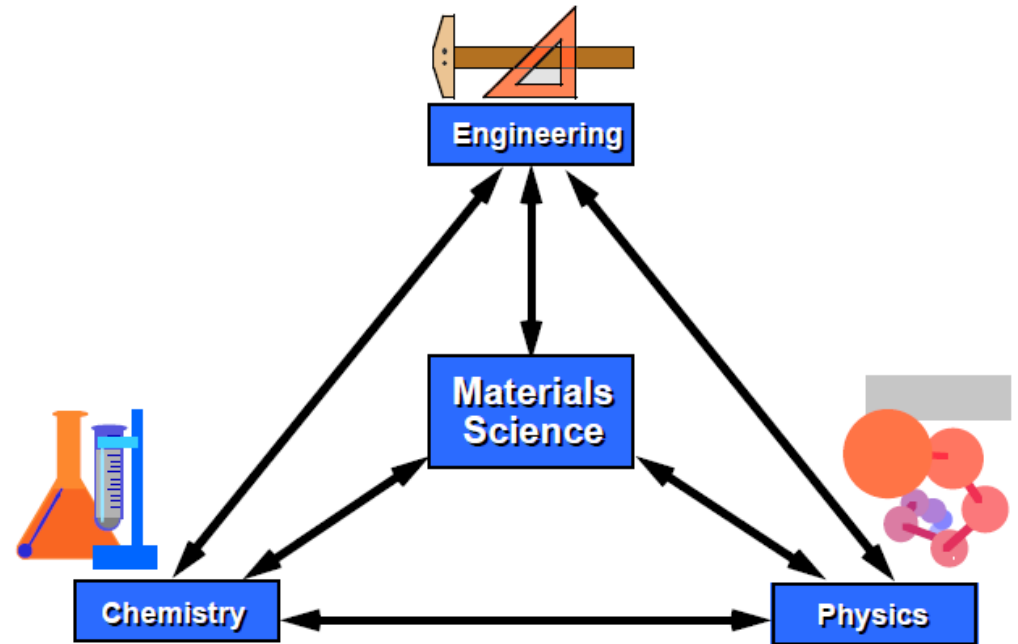
Importance of Study



the rubber seal or O-ring, had hardened overnight in freezing weather and failed when boosters ignited at launch

Materials Science

- Materials science combines with many areas of science and how materials science draws from chemistry, physics, and engineering to make better, more useful, and more economical and efficient stuff
- Materials Science – Investigating relationships that exist between the structure and properties of materials
- Materials Engineering is, on the basis of the structure-property correlations, designing or engineering the structure of a material to produce a pre-determined set of properties
- Materials science is an interdisciplinary field involving the properties of matter and its applications to various areas of science and engineering.



Materials Engineering & Technology - Tetrahedron

- Materials Science and Engineering is an interdisciplinary field concerned with inventing new materials and improving the previously known or existing materials by developing a deeper **understanding Structure – Property - Composition – Synthesis – Processing relationships**

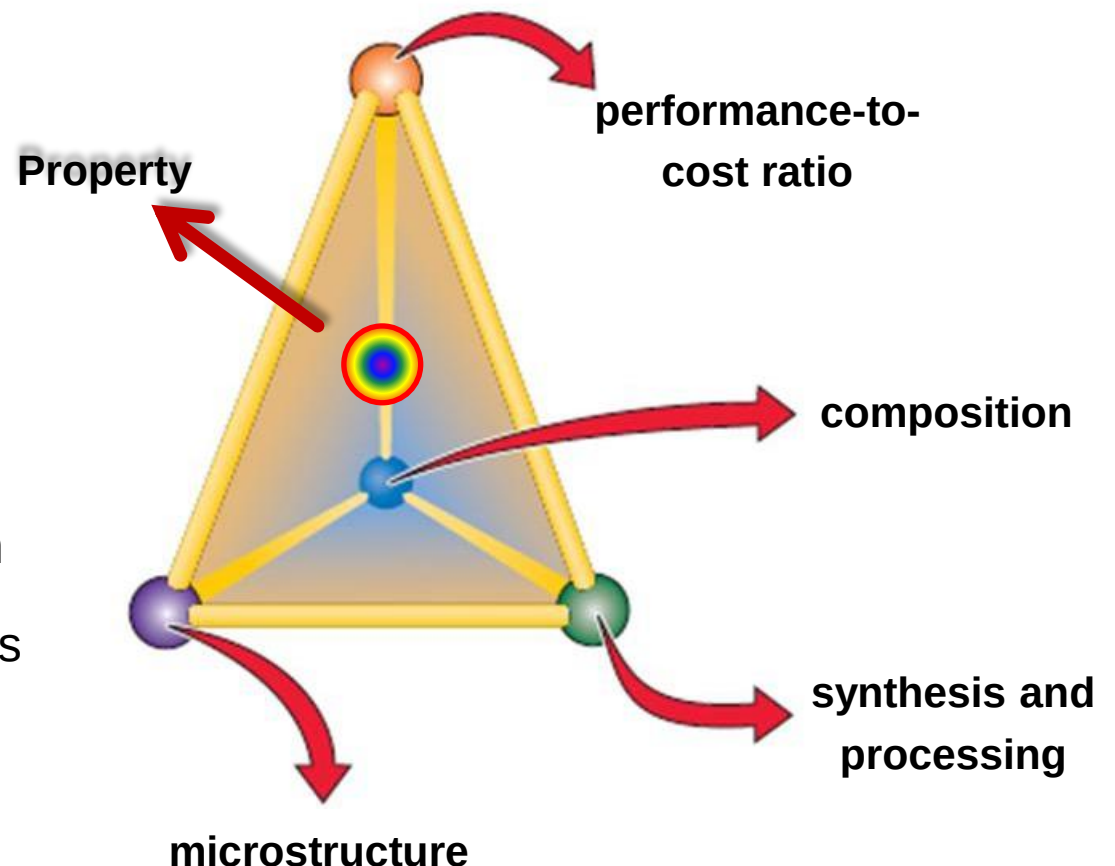
Tetrahedron Details

Composition

- Chemical make up of the material

Synthesis

- Refers to how materials are made from naturally occurring / man-made chemicals (i.e.) ores



Tetrahedron Details

Structure – Structure refers to the arrangement of a material's components from an atomic to a macro scale. Understanding the structure of a substance is key to understanding the state or condition of a material, information which is then correlated with the processing of the material in tandem with its properties., Macrostructure, Microstructure, Nanostructure, Crystal structure, Atomic structure

Processing - refers to the way in which a material is achieved.

- **Solidification Processing** - Most metals are formed by creating an alloy in the molten state, where it is relatively easy to mix the components. This process is also utilized for glasses and some polymers
- **Powder Processing** - Powder processing involves consolidation, or packing, of particulate to form a 'green body'. Densification follows, usually by sintering.
- **Deposition Processing** - Deposition processing modifies a surface chemically, usually by depositing a chemical vapor or ions onto a surface. It is used in semiconductor processing and for decorative or protective coating
- **Deformation Processing** - One of the most common processes is the deformation of a solid to create a desired shape.

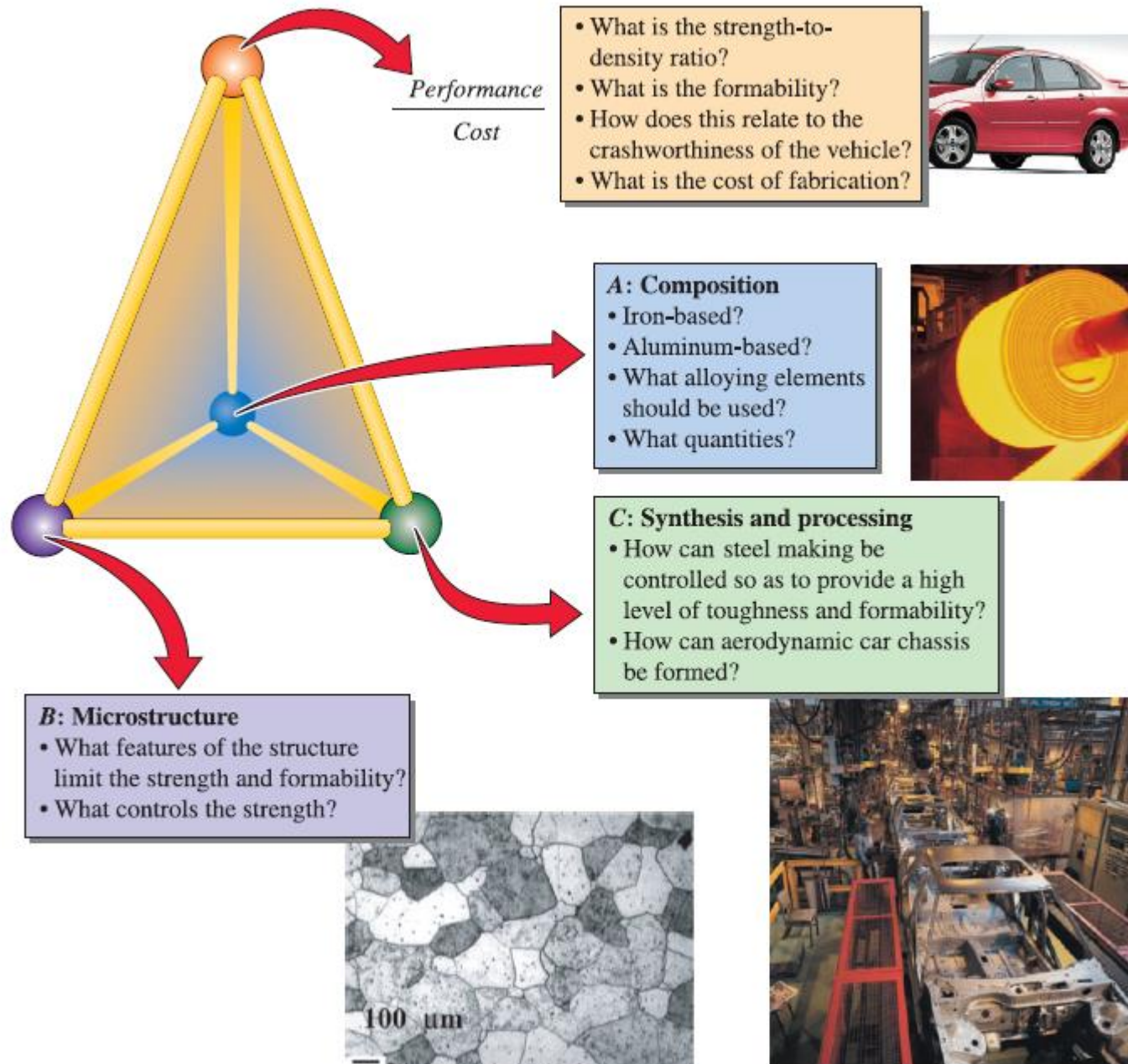
Tetrahedron Details

Properties - Does a material need to be strong and heat-resistant, yet lightweight? Is it possible to bring all the properties in one single material?

Whether you're talking about a fork or the space shuttle, products have specific requirements which necessitate the use of materials with unique properties

- **Mechanical Properties:** Tensile strength, fracture toughness, fatigue strength, creep strength, hardness, shock resistance
- **Electrical Properties:** Conductivity or resistivity, ionic conductivity, semiconductor conductivity (mobility of holes and electrons)
- **Magnetic Properties:** Magnetic susceptibility, Curie Temperature, Neel Temperature, saturation magnetization
- **Optical and Dielectric Properties:** Polarization, capacitance, permittivity, refractive index, absorption
- **Thermal Properties:** Coefficient of thermal expansion, heat capacity, thermal conductivity
- **Environmental Related Properties:** Corrosion behavior, wear behavior

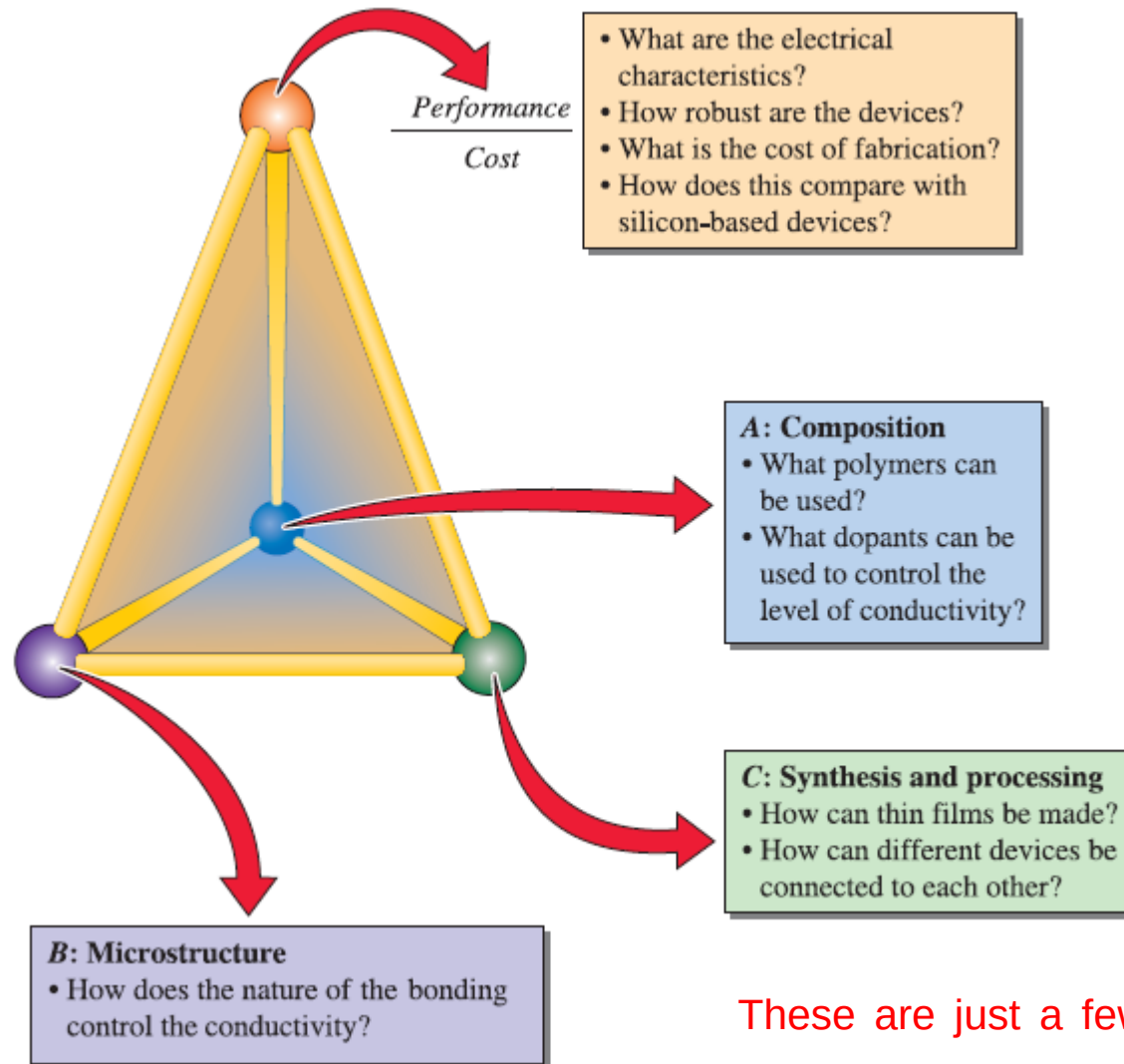
Tetrahedron Details - Application 1



Application to **sheet steels for automotive chassis.**

Note that the composition, microstructure, and synthesis-processing are all interconnected and affect the performance-to-cost ratio.

Tetrahedron Details – Application 2



- What are the relationships between the structure of polymers and their electrical properties?
- How can devices be made using these plastics?
- Will these devices be compatible with existing silicon chip technology?
- How robust are these devices?
- How will the performance and cost of these devices compare with traditional devices?

These are just a few of the factors that engineers and scientists must consider during the development, design, and manufacture of semiconducting polymer devices

Classification of Materials

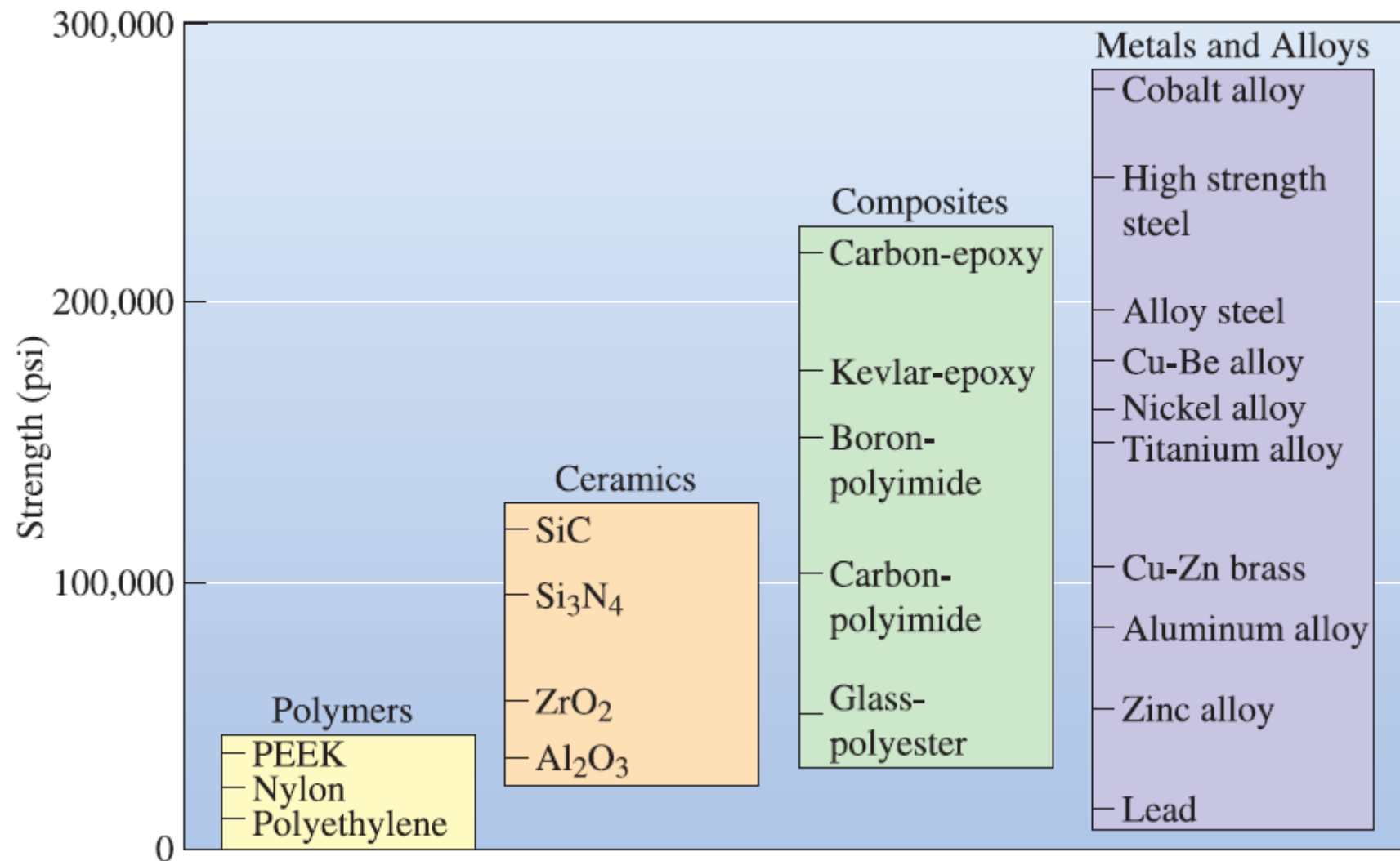
Generalized Classification

1. metals and alloys;
2. ceramics, glasses & glass-ceramics;
3. polymers (plastics);
4. semiconductors; and
5. composite materials.

TABLE 1-1 ■ *Representative examples, applications, and properties for each category of materials*

	Examples of Applications	Properties
Metals and Alloys		
Copper	Electrical conductor wire	High electrical conductivity, good formability
Gray cast iron	Automobile engine blocks	Castable, machinable, vibration-damping
Alloy steels	Wrenches, automobile chassis	Significantly strengthened by heat treatment
Ceramics and Glasses		
$\text{SiO}_2\text{--Na}_2\text{O--CaO}$	Window glass	Optically transparent, thermally insulating
Al_2O_3 , MgO, SiO_2	Refractories (i.e., heat-resistant lining of furnaces) for containing molten metal	Thermally insulating, withstand high temperatures, relatively inert to molten metal
Barium titanate	Capacitors for microelectronics	High ability to store charge
Silica	Optical fibers for information technology	Refractive index, low optical losses
Polymers		
Polyethylene	Food packaging	Easily formed into thin, flexible, airtight film
Epoxy	Encapsulation of integrated circuits	Electrically insulating and moisture-resistant
Phenolics	Adhesives for joining plies in plywood	Strong, moisture resistant
Semiconductors		
Silicon	Transistors and integrated circuits	Unique electrical behavior
GaAs	Optoelectronic systems	Converts electrical signals to light, lasers, laser diodes, etc.
Composites		
Graphite-epoxy	Aircraft components	High strength-to-weight ratio
Tungsten carbide-cobalt (WC-Co)	Carbide cutting tools for machining	High hardness, yet good shock resistance
Titanium-clad steel	Reactor vessels	Low cost and high strength of steel with the corrosion resistance of titanium

Representative strengths of various categories of materials



Metals & Alloys

Pure metals: are elements which comes from a particular area of the periodic table. Eg., copper in electrical wires and aluminum in cooking foil and beverage cans.

Metals are elements that generally have **good electrical and thermal conductivity**.

- Many metals have **high strength, high stiffness, and have good ductility and shock resistance**
- Some metals, such as iron, cobalt and nickel are magnetic. At extremely low temperatures, some metals and inter-metallic compounds become superconductors.
- Metal alloys contain more than one metallic element. Their properties can be changed by changing the elements present in the alloy. Eg., stainless steel, alloy of iron, nickel, and chromium; and gold jewelry which usually contains an alloy of gold and nickel, cadmium.
- Many metals and alloys have high densities and are used in applications which require a high **mass-to-volume ratio**.
- Some metal alloys, such as those based on **aluminum, have low densities and are used in aerospace applications for fuel economy**.

Ceramics

- inorganic crystalline materials; naturally occurring materials
- Outstanding properties : **Hard, brittle, high temperature resistance**
- Ceramics are used in the substrates that houses computer chips, capacitors and spark plugs
- Some ceramics such as silicon based ceramic barrier coatings show great potential for use in advanced, higher efficiency engines – (Source: NASA featured article on technology dated 29/03/11)
- Traditional ceramics are used to make bricks, refractories / abrasives
- Advanced ceramics offer higher strength, better wear & corrosion resistance, enhanced thermal shock
- Ceramics are used to make the cutting tools – **Boron Carbide, Boron Nitride; Grinding Wheels – SiC, Alumina**
- Structural clay products (bricks, sewer pipe, roofing and wall tile, flue linings, etc.)
- White-wares (dinnerware, floor and wall tile, electrical porcelain, etc.)
- Refractories (brick and monolithic products used in metal, glass, cements, ceramics, energy conversion, petroleum, and chemicals industries)

Ceramics and Advanced Ceramics

- Glasses (flat glass (windows), container glass (bottles), pressed and blown glass (dinnerware), glass fibers (home insulation), and advanced/specialty glass (optical fibers))
- Abrasives (natural garnet, diamond, etc.) and synthetic abrasives (silicon carbide, diamond, fused alumina, etc.) are used for grinding, cutting, polishing, lapping, or pressure blasting of materials)
- Cements (for roads, bridges, buildings, dams, and etc.)

Advanced ceramics

- Structural (wear resistant parts, cutting tools, and engine components)
- Electrical (capacitors, insulators, substrates, integrated circuit packages, piezo-electrics, magnets and superconductors)
- Coatings (engine components, cutting tools, and industrial wear parts)

Composites

Blending different properties of the material so as to get a single material with unique properties

- Composite material may be defined as 2 or more materials (phases/constituents) integrated to form a newer one
- The individual materials that make up composites are called constituents.
- Most composites have two constituent materials: **a binder or matrix**, and **a reinforcement**. The **reinforcement is usually much stronger and stiffer than the matrix**, and gives the composite its good properties.
- **A common example of a composite is concrete. It consists of a binder (cement) and a reinforcement (gravel). Adding another reinforcement (rebar) transforms concrete into a three-phase composite.**
- The matrix holds the reinforcements in an orderly pattern. Because the reinforcements are usually discontinuous, the matrix also helps to transfer load among the reinforcements; **Reinforcements basically come in three forms: particulate, discontinuous fiber, and continuous fiber.**

Composites

Matrix and Reinforcements:

- Matrix materials Polymers, Metals, Ceramics and
- Reinforcement: fibers Glass, Carbon, Organic Boron, Ceramic, Metallic,

Material	Characteristics
Fibers	
Glass	High strength, low stiffness, high density; lowest cost; E (calcium aluminoborosilicate) and S (magnesia-aluminosilicate) types commonly used.
Graphite	Available as high-modulus or high-strength; low cost; less dense than glass.
Boron	High strength and stiffness; highest density; highest cost; has tungsten filament at its center.
Aramids (Kevlar)	Highest strength-to-weight ratio of all fibers; high cost.
Other fibers	Nylon, silicon carbide, silicon nitride, aluminum oxide, boron carbide, boron nitride, tantalum carbide, steel, tungsten, molybdenum.
Matrix materials	
Thermosets	Epoxy and polyester, with the former most commonly used; others are phenolics, fluorocarbons, polyethersulfone, silicon, and polyimides.
Thermoplastics	Polyetheretherketone; tougher than thermosets but lower resistance to temperature.
Metals	Aluminum, aluminum-lithium, magnesium, and titanium; fibers are graphite, aluminum oxide, silicon carbide, and boron.
Ceramics	Silicon carbide, silicon nitride, aluminum oxide, and mullite; fibers are various ceramics.

Composites

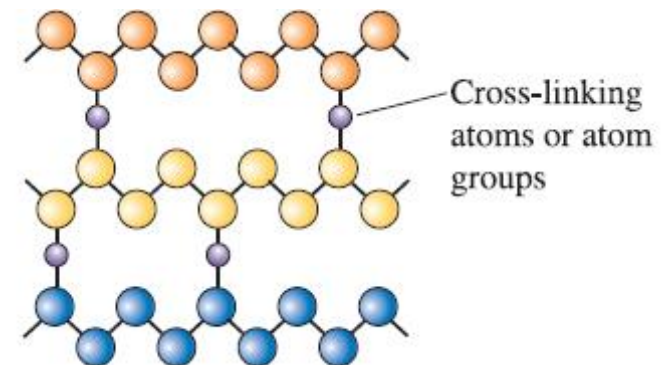
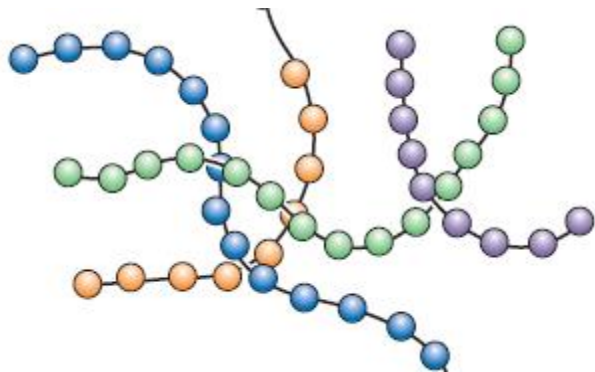
- Glass reinforced composites are the most desired materials as a result of advanced technology that has gone beyond the design and application
- Graphite is a widely available economical reinforcement material with high stiffness, high modulus, high strength and high theoretical efficiency
- The first structural composite aircraft components, which were introduced during 1950-60, were made from glass fibre reinforced plastics. These components included the fin and the rudder of Grumman E-2A, helicopter canopies, frames, radomes, fairings, rotor blades, etc.
- Due to high strength and stiffness combined with low density, composites like Boron Fibre Reinforced Plastics (BFRP) and Carbon Fibre Reinforced Plastics (CFRP) were preferred instead of aluminium for high performance aircraft structures. For lightly loaded structures, Aramid Fibre Reinforced Plastics (AFRP) which possess low density, have been used in versatile applications

Polymers

- Polymers are typically organic materials.
- They have lower strength; but high strength to weight ratio.
- Not suitable for high temperature applications
- Many polymers have good resistance to corrosion and good electrical conductivity
- Polymers have thousands of applications ranging from bullet proof vests, compact discs, ropes and LCDs.
- Polymers are of two types – Thermosetting & Thermoplastics

Polymers

- **Thermoplastic polymers** are normally produced in one step and then made into products in a subsequent process.
- They become soft and formable when heated. When cooled significantly below their softening point they again become rigid and usable as a formed article.
- This type of polymer can be readily recycled because each time it is reheated it can again be reshaped or formed into a new article
- **Thermosetting polymers** are normally produced and formed in the same step.
- Upon heating, thermosetting polymers will become soft, but cannot be shaped or formed to any great extent, and will definitely not flow.
- These forms have very strong bonds between the different chains.
- This makes it almost impossible for the chains to slide past each other and result in plastics that are both hard and brittle.



Semiconductors

- relatively small group of elements and compounds has an important electrical property, **semi-conduction**, in which they are **neither good electrical conductors nor good electrical insulators**.
- Instead, their ability to conduct electricity is intermediate.
- These materials are called **semiconductors**, and in general, they do not fit into any of the four structural materials categories based on atomic bonding.
- Si, Ge, GaAs are the best examples for Semiconductors

Functional Classification of Materials



Functional Classification of Materials

Aerospace

- Wood ---- Steel ---- Al alloys ----- Composites
- NASA's space shuttle make use of Al powders for booster rockets tiles
- Al alloys, Plastics, Silica – for making space shuttle tiles

Bio-Medical

- Bones & teeth are made up of naturally occurring ceramic – Hydroxyapatite
- Plastics, Ti alloys, Austenitic stainless steels, composites are used for making artificial organs, prosthetic limbs, bone replacement parts, cardiovascular stents etc.
- Ultrasonic imaging systems make use of PZT (lead zirconium titanate)
- Magnetic Resonance Imaging (MRI) – makes use of Ni-Sn based superconductor

Functional Classification of Materials

Electronic Materials

- Cu, Al & other metals are used as conductors in power transmission
- BaTiO_3 & Ta_2O_5 are used for making ceramic capacitors
- Si, GaAs, Ge, W and other conducting polymers are used as electronic materials

Energy & Environmental

- Nuclear industry make use of materials such as Uranium oxide & Plutonium as fuels
- Glasses & Stainless Steels are used handling the nuclear radioactive wastages
- Batteries and fuel cells make use of many ceramic materials such as Zirconia & Polymers
- Oil & Petroleum industry widely utilizes Zeolites, Alumina & other materials as catalyst substrates include Pt/Rh, Pt
- Solar power is generated using materials such as Amorphous Silicon

Functional Classification of Materials

Magnetic

- Computer hard disks & audio-video cassettes make use of a combination of ceramic, metallic and polymeric materials
- $\gamma\text{-Fe}_2\text{O}_3$ are deposited on to a polymeric base to make the cassettes
- Computer hard disks are made up of alloy consist of Co-Pt-Ta-Cr
- Steels based on Fe & Si are widely used for making transformer cores.

Photonic

- Silica is widely used for making optical fibers
- YAG & Al_2O_3 are used for making lasers
- Amorphous Silicon is used to make Photovoltaic Modules
- Polymers are used in making LCDs

Functional Classification of Materials

Smart Material

- PZT & Shape memory alloys
- When properly processed PZT can be subjected to stress, a voltage is generated. This effect can be used to make such devices as spark generators for gas grills & sensors that can detect underwater objects
- MR Fluids (Magneto-rheological fluids) - are being widely used in the suspension system of the automobiles
 - Magneto-rheological fluids (MRF) consist of magnetisable solid particles (mostly iron) in a carrier fluid. Upon applying an external magnetic field, the particles are magnetised and form chains along the lines of the magnetic flux. This causes the MRF to change from a liquid to a solid state within milliseconds
- Automatic dimming mirrors and photo-chromic glasses make use of smart materials

Examination of Structure

- Structure tells the arrangement of material's components from macro scale to atomic level
 - Macrostructure
 - Microstructure
 - Nanostructure
 - SRO & LRO
 - Crystal Structure
 - Atomic Structure

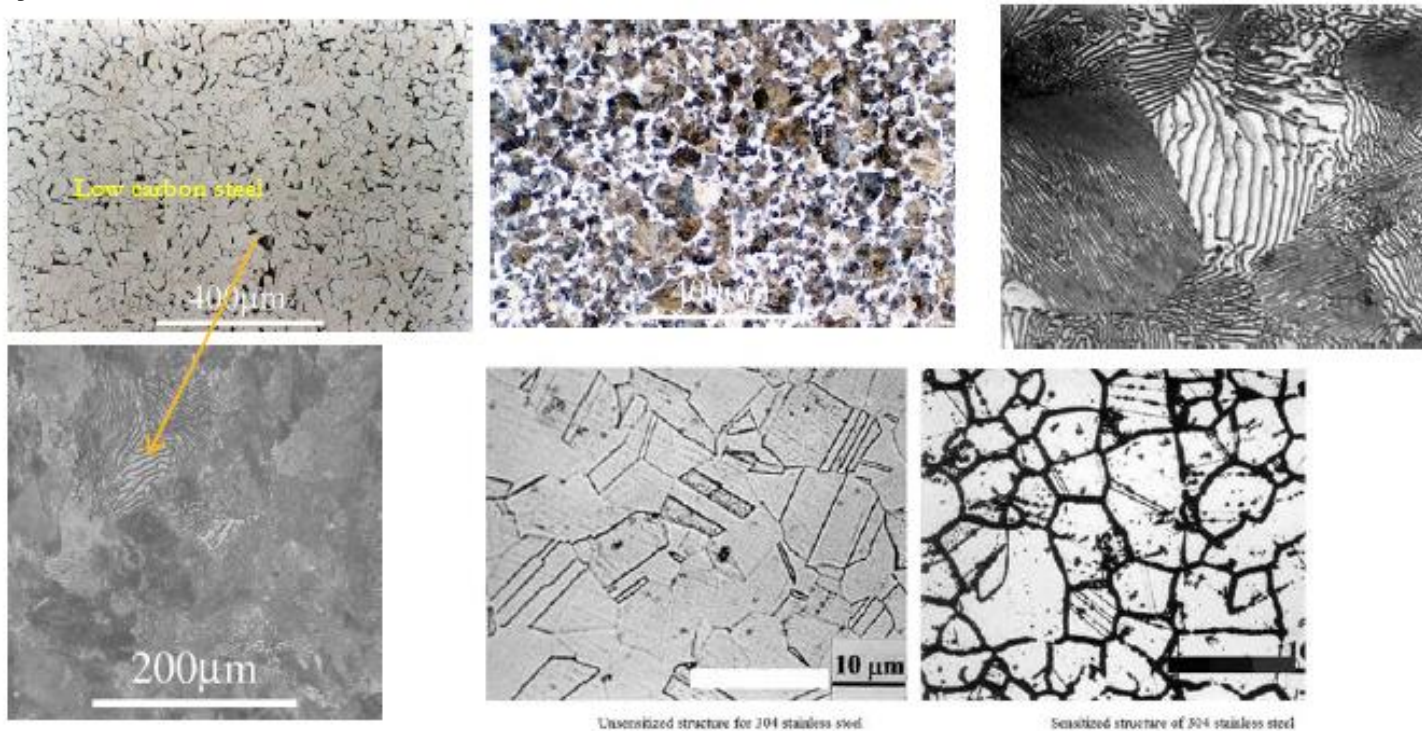
Macrostructure

- Studies / features contributes this structure include coating thickness, external porosity, flaws and cracks
- Visual Examination or low magnification microscope with 5x – 20x magnification



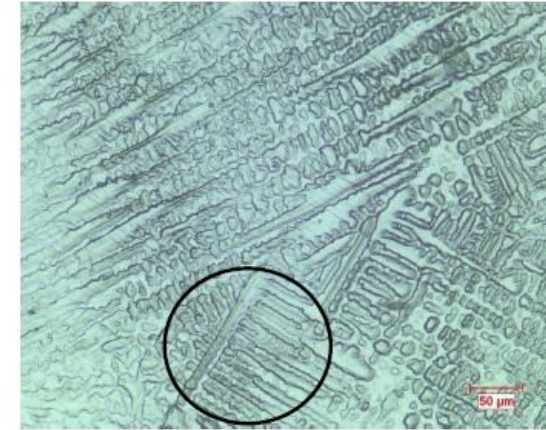
Microstructure

- Study of material at its microscopic level ranging in microns to some nanometers
- Features include grain size, orientation of grains, amount of elements in the matrix
- Internal morphology of the material can be studied
- Lower end microscopes – 5x – 200x
- Higher end microscopes - >500x
- SEM analysis reveal the date to about 1,00,000 – 5,00,000 x

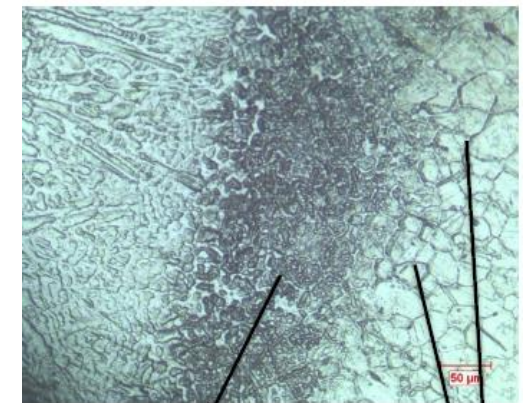


Observations from Microstructure studies

- A grain is the portion of the material within which the arrangement of atoms are nearly identical
- Length Scale Range – It is the distance between the atoms
- When a metal solidifies from the molten state, millions of tiny crystals start to grow.
- The longer the metal takes to cool the larger the crystals grow.
- These crystals form the grains in the solid metal.
- Each grain is a distinct crystal with its own orientation.

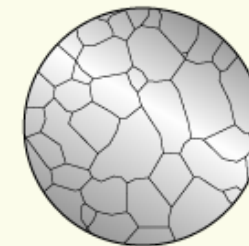
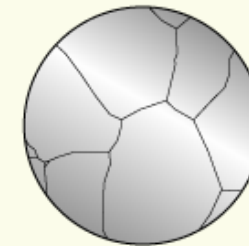
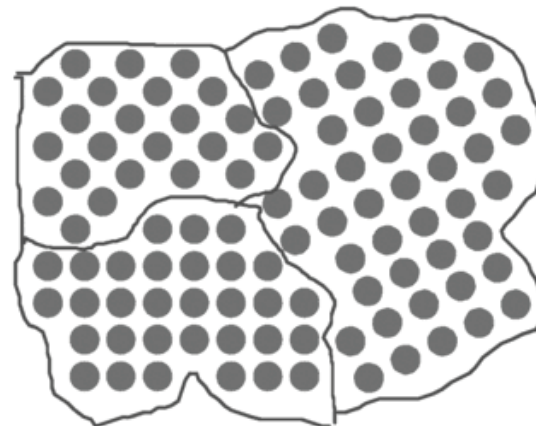
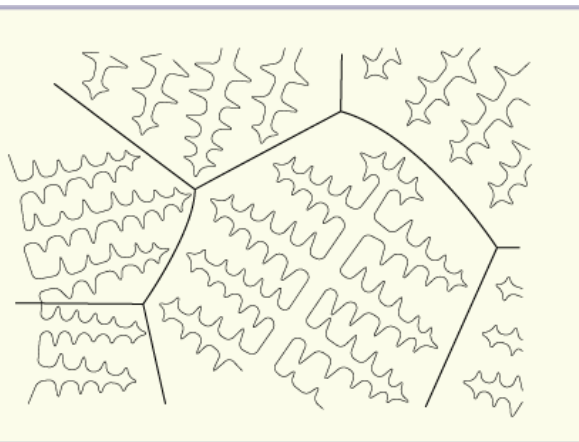


Dendritic growth - Solidification

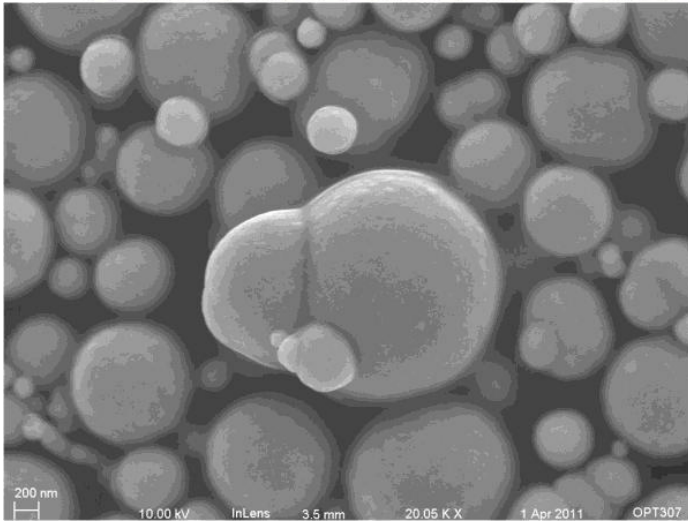


Segregation at the HAZ

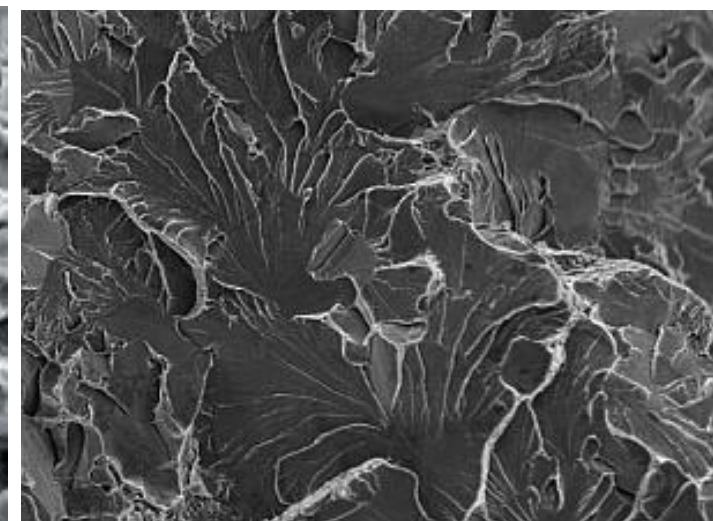
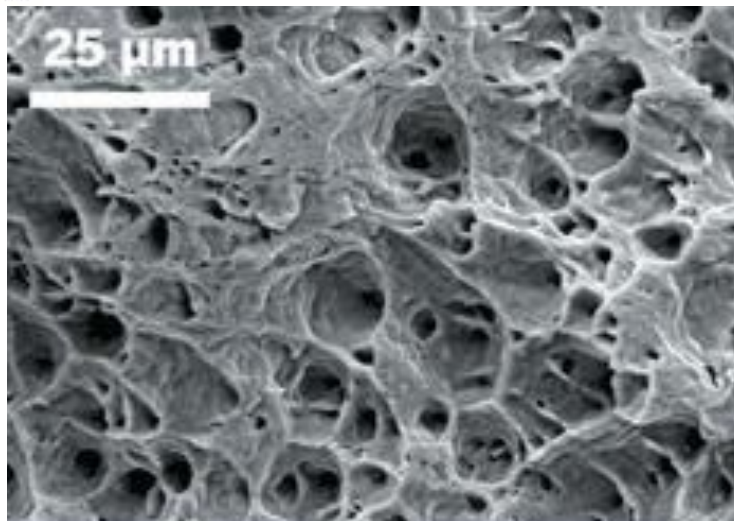
Grain boundary thickening



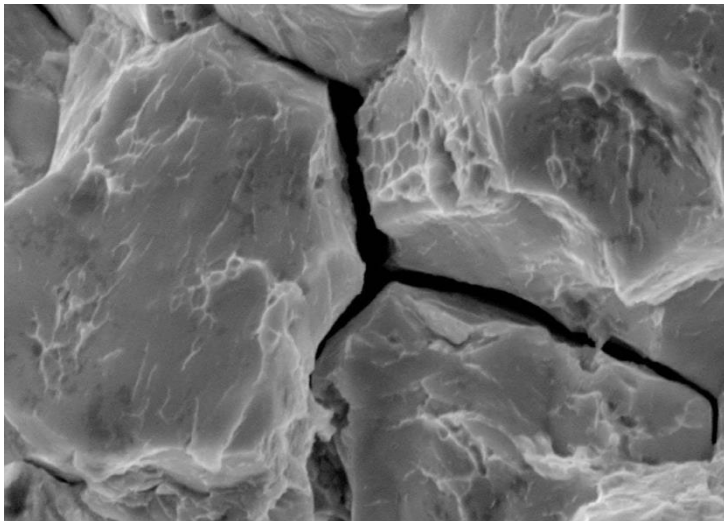
Hi Resolution Microscopy



Coating



SEM Fractography - Void Formation



Hydrogen Embrittlement



Corrosion Attack

Nanostructure

- Nano-crystalline materials are single- or multi-phase polycrystalline solids with a grain size of a few nanometers (10 Å), typically less than 100 nm
- Nanostructures formed chemically under ambient conditions can also be found in natural biological systems from seashells to bone and teeth in the human body
- In comparison to the coarse-grained materials, nano-crystalline materials show **higher strength and hardness, enhanced diffusivity, and superior soft and hard magnetic properties.**
- **It is often stated that as grain size moves to nanoscale, metals get stronger and harder (and more brittle) while ceramics become more ductile (and malleable).** This is an approximation and in fact the reality is more complex and dependent on what part of nano-scale the grain sizes are in
- **Nano-crystalline silicon** has properties in addition to electroluminescence (of interest for semiconductor laser applications) such as photoluminescence and thermally induced acoustic emission
- **Nano-sized particles (approx. 5nm) of Fe_2O_3 are used in liquid magnets/ferro-fluids – These are used as cooling medium for loud speakers**

Nanostructure

- Nanostructured materials have been synthesized in recent years by methods including *inert gas condensation, mechanical alloying, spray conversion processing, severe plastic deformation, electrodeposition from the melt, physical vapor deposition, chemical vapor processing, co-precipitation, sol-gel processing, sliding wear, spark erosion, plasma processing, auto-ignition, laser ablation, hydrothermal pyrolysis, thermophoretic forced flux system, quenching the melt under high pressure, biological templating, sonochemical synthesis, and devitrification of amorphous phases*
- High hardnesses and yield strength values are observed for nanocrystalline materials.
- Superplasticity has been observed at low temperatures ($0.38T_m$) for nanocrystalline nickel and nickel aluminide samples.
- Currently, bulk nanostructured soft magnetic iron based alloys and WC–Co nanocomposites have found industrial uses
- Nanocrystalline coatings deposited by laser plasma discharge increased the life of ZnS samples more than five times against abrasion/erosion/rain water corrosion/impact damage

Short Range & Long Range Order

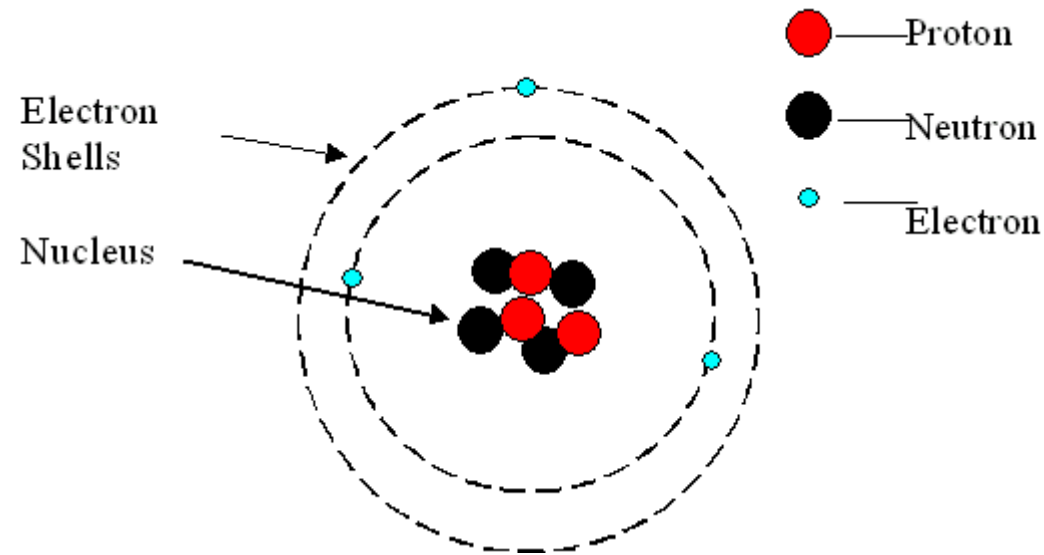
If the arrangement of atoms are regular, periodic, continuous and in an orderly fashion and also if the regularity & repeatability is achieved for a longer distances, throughout the entire volume then the materials termed to have ***long range order***

If the arrangement of atoms are irregular, discontinuous and in disordered fashion, then these materials will have the orderliness only for a short distances, these are refereed to as ***Short range order***

- Crystalline materials are usually possessing LRO whereas the amorphous solids possess SRO;
- Single-crystal materials feature long-range order throughout the entire piece of material while poly-crystalline materials feature long-range order only within limited grains.

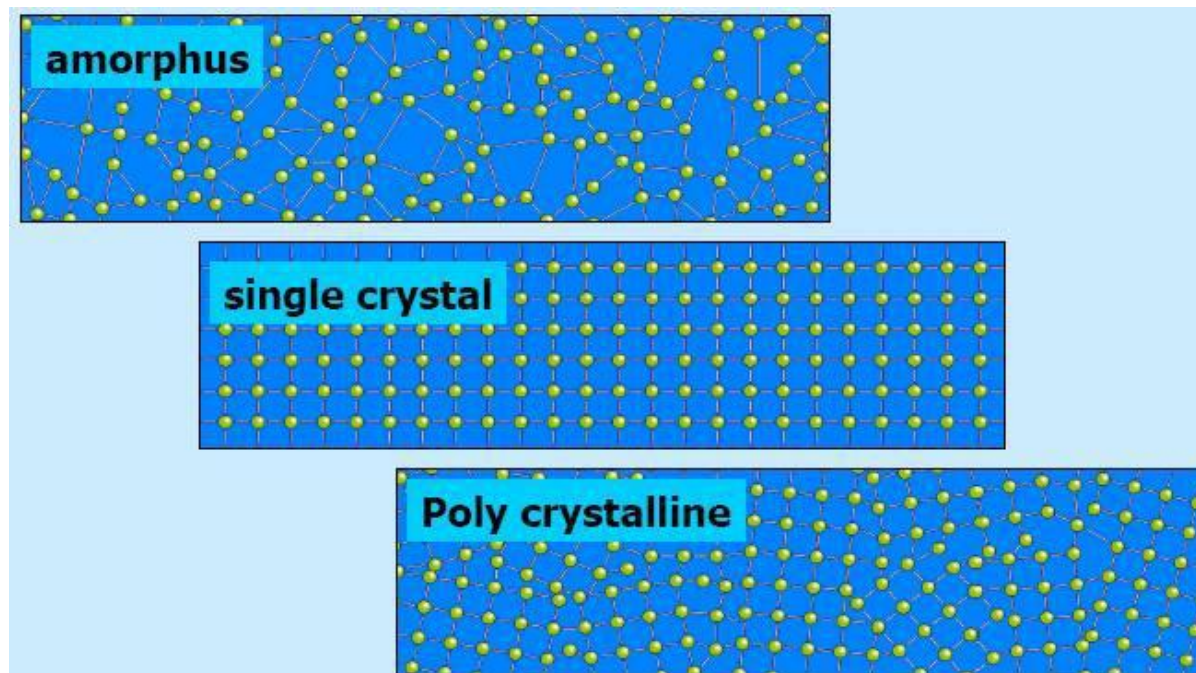
Atomic Structure

- Features for how the atomic bonds lead to the different atomic and ionic arrangements
- It includes all atoms and the way how they are arranged
- The insights gained by understanding the atomic structure and bonding configuration of atoms/molecules are essential for the selection of engineering materials



Crystalline Materials

- A material has LRO if it exhibits the order over long distances throughout its entire volume
- The bond length will be uniform, repetitive and properly fashioned
- Crystalline structure is important because it contributes a lot to decide and alter the properties of a material. For example, it is easier for planes of atoms to slide by each other if those planes are closely packed.
- Crystals can be classified into (i) Single Crystalline and (ii) Poly-crystalline materials



Single Crystalline Materials

- A single /mono-crystal is a crystalline solid in which the crystal lattice of the entire sample is continuous and unbroken to the edges of the sample with no grain boundaries
- The absence of defects associated with grain boundaries can give mono-crystals with unique properties in terms of mechanical, optical and electrical

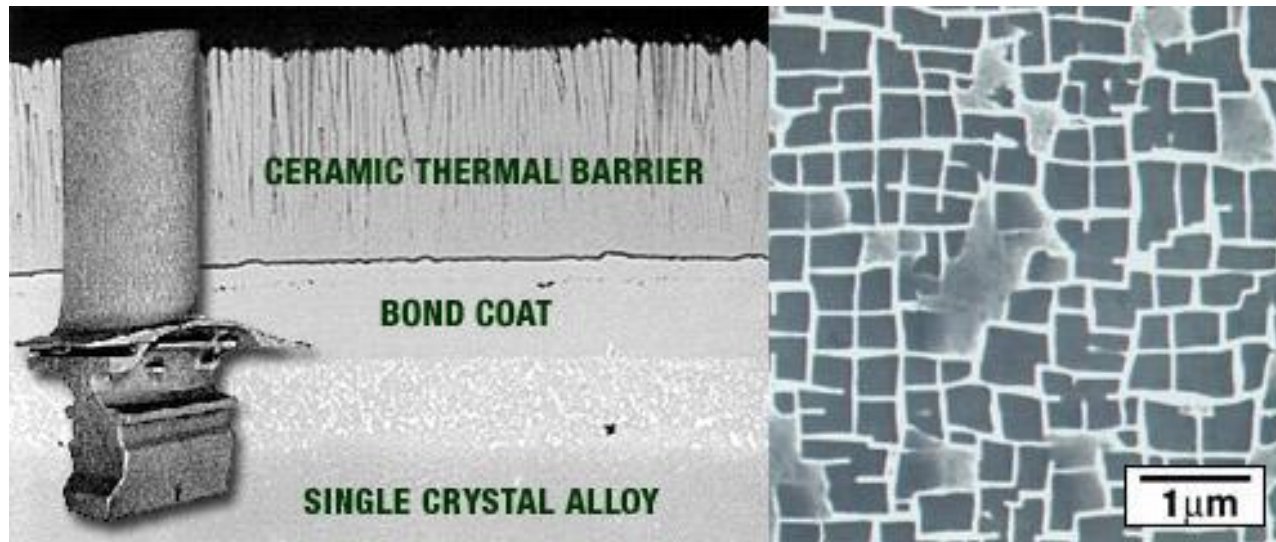
Typical Uses

- Mono-crystals of Sapphire & other materials are used for making lasers and non-linear optics
- Materials used to fabricate single-crystal piezoelectric elements include
 - lead magnesium niobate / lead titanate (PMN-PT),
 - lead zirconate niobate /lead titanate (PZN-PT),
 - lithium niobate (LiNbO_3), lithium niobate with dopants,
 - lithium tetraborate ($\text{Li}_2\text{B}_4\text{O}_7$), and quartz.
 - Barium titanate (BaTiO_3) is a potential non-lead source of piezoelectric crystals for low temperature and room temperature applications

Single Crystalline Materials

- Other applications for lithium tetraborate crystals include bulk acoustic wave (BAW) devices, pagers, cordless and cellular telephones, and data communication devices.
- Applications for quartz crystals include timing mechanisms for watches and clocks and delay lines for electrical circuits.
- The performance of a single-crystal element depends on the direction in which the raw crystal is cut
- Single Crystals of Cu has better conductivity than the polycrystalline one
- Mono-crystals of fluorite are used in the fabrication of refractory telescopes
- NASA utilized single crystal superalloys which offer improved stress rupture life, low and high cycle fatigue life; however the mechanical properties of super conducting materials are highly anisotropic

Typical Application (NASA Source)



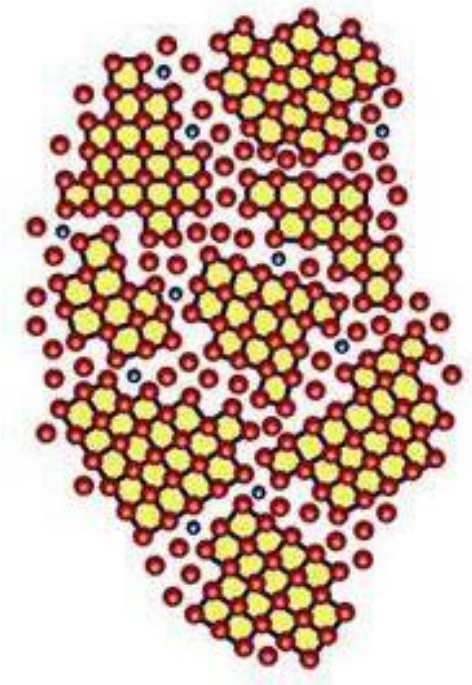
High Pressure Turbine Blades consist of a 3-part system:

- The single crystal Ni substrate,
- The bond coat/environmental barrier, and
- The thermal barrier coating. The image above shows a high pressure turbine blade (left), structure of the blade (center) and a magnified view of a single crystal alloy (right).

Reference : http://www.grc.nasa.gov/StructuresMaterials/AdvMet/research/turbine_blades.html

Polycrystalline Materials

- A poly-crystalline material is composed of numerous, tiny crystals called crystallites; usually these have grain boundaries
- Poly-crystalline materials contain regions where the different grains meet is called as grain boundary
- Size of the grains can range from nanometer to being visible through naked eye
- Grain boundaries profoundly affect the mechanical and electrical properties of polycrystalline materials
- Depending on the size of the grain, the properties will vary
- Grains are classified based on the size (i) Coarse Grain (ii) Fine grain
- Grain size is normally quantified by a numbering system. Coarse 1-5 and fine 5-8.
- The number is derived from the formula $n=2^{N-1}$, where n = the number of grains per square inch at 100X magnification, N = ASTM Grain Size number
- Fine grained materials offer higher tensile strength and ductility under ambient conditions



Liquid Crystals

- These are the polymeric materials that have a special type of order
- LC Polymers behave as amorphous materials (liquid like) in one state.
- However when an external stimulus (such as an electric field/ a temp. change) is applied, the polymer molecules undergo alignment & forms small regions that are crystalline and hence the name “liquid crystals”

Amorphous Materials

- They have short range atomic arrangement of atoms/ions show a particular order relatively for a short distances
- Amorphous materials are often prepared by rapidly cooling the molten metal (such as glass).
- The cooling reduces the mobility of material's molecules before they can pack into a more thermodynamically stable state
- Amorphous materials can also be produced by additives which interfere with the ability of primary constituents to crystallize
- Eg. Addition of soda to SiO_2 results in window glass & addition of glycols to water into vitrified solid